INCH-POUND
MIL-M-24784/10(SH)
6 September 1994
SUPERSEDING
MIL-M-24571(SH)
25 October 1979

ASSOCIATED DETAIL SPECIFICATION

MANUAL, TRAINING AID BOOKLET (TAB) REQUIREMENTS FOR NAVAL SHIPS SYSTEMS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sets forth requirements for the preparation of Training Aid Booklets (TABs) for use on naval ships (see 6.1).

1.1.1 TAB coverage. The TAB is a schematic and pictorial representation of systems and equipment installed aboard a specific ship. TAB coverage does not include maintenance information, nor does the TAB coverage include the internal details of nomenclatured or otherwise uniquely identified systems or equipments documented by other technical manuals supplied aboard ship.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-M-24784 - Manual, Technical: General Acquisition and Development Requirements.
MIL-M-24784/1 - Manual, Technical Change Package Requirements.
MIL-M-24784/2 - Manual, Technical Revision Requirements.

STANDARD

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings, Specifications, Standards, and in Technical Documents.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

DEPARTMENT OF DEFENSE

DoD 5200.1-R - Information Security Program Regulation.
DoD 5220.22-M - Industrial Security Manual for Safeguarding Classified Information.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-5001049 - Piping System Symbols and Abbreviations.

(Application for copies should be addressed to the Commander, Portsmouth Naval Shipyard, Naval Engineering Drawing Support Activity, Code 280.2, Portsmouth, NH 03804-5000.)

2.2 Non-Government publications. The following non-Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
ANSI Y14.15 - Electrical and Electronics Diagram (Including Supplements Y14.15a and Y14.15b).
ANSI Y32.4 - Graphic Symbols for Plumbing Fixtures for Diagrams Used in Architecture and Building Construction.
ANSI Y32.10 - Graphic Symbols for Fluid Power Diagrams.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
IEEE Std 315A-86 - Graphic Symbols for Electrical and Electronics Diagrams.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017 or from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications), the text of this specification shall take precedence. When a generic requirement is in conflict with a unique requirement from the governing associated detail specification, the unique requirement shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Security classifications. The overall classification assigned to a technical manual shall agree with the highest classification assigned to any portion within and shall be placed conspicuously at the top and bottom of the Cover/Title page or abbreviated title. Security classification markings shall be in accordance with DoD 5220.22-M and 5200.1-R and shall be placed in accordance with DoD 5220.22-M and 5200.1-R. The security markings shall be the same as for other title pages.

3.2 Deliverable products and data items. The deliverable products and data items shall be specified in the Contract Data Requirements List (CDRL) and shall conform to the requirements specified in MIL-M-24784 and herein. Data items shall include the following, as applicable:

(a) Outline or book plan.
(b) Review draft copy (RDC).
(c) Preliminary technical manual (PTM).
(d) Final reproducible copy (FRC) (see 3.2.1).
(e) Supplements.
(f) Changes (3.2.2).
(g) Revisions (3.2.3).
(h) Management data (see 3.2.4).

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3.2.1.4 Running feet.

3.2.1.4.1 Page number positioning. All page numbers shall be located at the lower outside of the page, approximately 1/4 inch from the bottom and 1/4 inch from the outside edge.

3.2.1.5 Page size. Unless otherwise specified (see 6.2), the page size for all preliminary and final TABs shall be 11 by 8 1/2 inches with a maximum image area of 9 1/2 by 7 1/2 inches (see figure 10-1). Alternate page size, when so specified, shall be 4-1/4 by 11 inches.

3.2.2 Change packages. When specified (see 6.2), change packages shall be in accordance with MIL-M-24784/1.

3.2.3 Revisions. When specified (see 6.2), revisions shall be in accordance with MIL-M-24784/2.

3.2.4 Management data. Unless otherwise specified (see 6.2), the following quality assurance data items shall be provided in accordance with MIL-M-24784:

(a) Technical Manual Quality Assurance Program Plan.
(b) Validation Plan.
(c) Validation Certification.
(d) Technical Manual Evaluation Records.
(e) Verification Plan.
(f) Technical Manual Verification Incorporation Certification.

3.3 Arrangement. Unless otherwise specified, the manual shall be arranged in a standardized format (that is, front matter, technical content, appendices, glossaries, indexes and back matter) and appropriately divided by volume, part, chapter and section in accordance with the requirements of MIL-M-24784, Appendix C and the following.

3.3.1 Division of coverage. The TAB shall consist of at least two unclassified volumes. A third volume (supplement) shall be used when classified information must be included in the TAB coverage. The division of coverage shall be as follows:

(a) Volume 1 - Piping systems.
(b) Volume 2 - Electrical and electronic systems.
(c) Volume 3 - Supplement (classified data applicable to both volumes).

3.3.2 Volume and part limitation. Volumes, or parts thereof, shall not exceed 500 pages (250 sheets).

3.3.3 Cover or title page. The cover or title page shall be prepared in a horizontal (landscape) format as shown in MIL-M-24784, Appendix C.

3.4 Format and preparation instructions.

3.4.1 Writing style. Unless otherwise specified, the style of writing, level of writing, readability, and referencing shall be in accordance with the requirements of MIL-M-24784, Appendix B.

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3.4.2 Safety precautions. Unless otherwise specified, safety and health warnings, cautions and notes shall be in accordance with the requirements of MIL-M-24784, Appendix D.

3.4.3 Tabular material. Unless otherwise specified, the preparation of tables shall be in accordance with the requirements of MIL-M-24784, Appendix E.

3.4.4 Graphics. Unless otherwise specified, the preparation of illustrations, drawings, diagrams and sketches shall be in accordance with the requirements of MIL-M-24784, Appendix F.

3.4.4.1 Illustration titles. Titles shall be located adjacent to the page number. The primary source document from which the illustration was developed shall be identified immediately below the illustration title in the manner shown in figures 10-1 and 10-2.

3.4.4.2 Line widths. Line artwork shall be prepared with line widths of sufficient size and strength to reproduce a minimum final line width, without dropout (when reduced), of 0.01 inch.

3.4.4.3 Piping illustrations. Pipe sizes shall be indicated by line width. A code to indicate the pipe size scale shall appear on the illustration. Recommended pipe size scale for 2.5 times oversized artwork is shown on figure 10-4.

3.4.4.4 Electrical system illustrations. Electrical distribution wire and cable sizes shall be annotated on the illustration. Bus bars shall be depicted in rectangular form and identified accordingly.

3.4.4.5 Illustration size. All original artwork shall be prepared for reproduction at final sizes as indicated in table II. Original artwork shall not exceed the maximum size indicated.

TABLE II. Illustration sizes.

Maximum original	Final	Final format (printed page)
23 3/4 by 18 3/4	9 1/2 by 7 1/2	Single illustration page (see figure 10-2)
23 3/4 by 8 3/4	9 1/2 by 3 1/2	Double illustration page (see figure 10-3)

3.4.4.6 Multiple or parallel installations. Only one system shall be illustrated when multiple or parallel systems or components are installed in the ship. A note shall be added to the illustration to indicate the existence of the system(s) or component(s) not shown.

3.4.4.7 Color in illustrations. When specified (see 6.2), only the Pantone PMS colors and variations of use identified in table III shall be used for TABs. The color-coded drawing copies shall identify each color by number and variation of use.

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TABLE III. Printing colors and variations.

Color	Pantone PMS number	Acceptable for use as			
		Solid area	Solid thin line	40 percent screened area	40 percent screened line
Red	185	Yes	Yes	Yes	Yes
Yellow	108	Yes	Yes	No	No
Green	354	Yes	Yes	Yes	Yes
Blue	299	Yes	Yes	Yes	Yes
Orange	151	Yes	Yes	No	No
Purple	253	Yes	Yes	No	No
Black	---	Yes	Yes	Yes	Yes

3.4.5 Numbering. Unless otherwise specified in the contract or order, the numbering of paragraphs, procedures, divisions, issues, changes, illustrations, tables, and publications shall be in accordance with the requirements of MIL-M-24784, Appendix G and the following.

3.4.5.1 Page and illustration numbering.

3.4.5.1.1 Front matter. Numbering of front matter shall be by volume and lower case consecutive Roman numerals (for example, 1-i, 1-ii, and so forth).

3.4.5.1.2 Text and illustrations. The content pages and illustrations shall be numbered by volume and consecutive Arabic numbers (for example, 1-1, 1-2, and so forth).

3.5 Technical content. The technical content of each volume of the TAB shall be as specified in 3.5.1 through 3.5.6 and shall be arranged as follows:

(a) General information (see 3.5.1).
(b) List of abbreviations (see 3.5.2).
(c) List of symbols (see 3.5.3).
(d) System and component designation tables (see 3.5.4).
(e) Perspective and plan views (see 3.5.5).
(f) System and component diagrams (see 3.5.6).

3.5.1 General information. The general information page(s) shall provide as a minimum the following data:

(a) Full title and Government identification numbers of all volumes and separately bound parts of volumes comprising the TAB (including supplements).
(b) Explanation of the purpose and scope of the TAB.
(c) Explicit applicability of the TAB.
(d) A brief synopsis of the content of each volume.
(e) A brief description of the graphic techniques used in the TAB.
(f) A statement on reporting of errors (see MIL-M-24784).

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3.5.1.1 Use of general information page(s). The general information page(s) shall be included in each volume of the TAB, and shall be identical.

3.5.1.2 Multipart volumes. When multiple parts are used for Volume 1 or Volume 2 (or both) of the TAB, the general information page(s) shall be included only in Part 1 of each volume.

3.5.2 List of abbreviations. A complete listing of abbreviations and their meaning shall be included in each volume immediately following the general information page(s). For multipart volumes, the list of abbreviations shall appear in Part 1 only.

3.5.2.1 Nonstandard abbreviations. The use of nonstandard abbreviations shall be held to a minimum. Nonstandard abbreviations shall be used only when no authorized equivalent exists in MIL-STD-12. When used, nonstandard abbreviations shall not be in conflict with a MIL-STD-12 abbreviation.

3.5.3 List of symbols. A list of symbols shall be included in each volume (Part 1 only, on multipart volumes) of the TAB, immediately following the list of abbreviations. The listing for each volume shall be limited to symbols used in the applicable volume but shall be inclusive of all symbols used therein.

3.5.3.1 Symbols. Graphic symbols for use on TAB diagrams shall be in accordance with the following:

(a) Electrical and electronic - ANSI Y14.15 and IEEE 315A-86.
(b) Plumbing - ANSI Y32.4.
(c) Piping - Drawing 803-5001049.
(d) Fluid power - ANSI Y32.10.

3.5.3.1.1 Nonstandard symbols. The use of nonstandard symbols shall be held to a minimum. Nonstandard symbols shall be used only when no authorized equivalent exists in the documents specified above. Where nonstandard symbols are used, explanations shall be provided. When used, nonstandard symbols shall not be in conflict with the above documents.

3.5.4 System and component designation tables. Data shall be included in tabular form to explain the use of color codes, piping system designators, location identification numbers for controls and instruments, switchboard and circuit breaker identifications, and so forth, and shall be arranged in logical order immediately following the list of symbols. Any other data that will enable users to better understand the TAB shall be presented in additional tables or by suitable notes positioned near the system or component illustrations to which they apply.

3.5.5 Perspective and plan views. Perspective and plan views (see figure 10-5) shall be included in the TAB to illustrate such specific aspects of the ship as:

(a) Antenna and mast arrangements.
(b) Exterior installations including superstructure details and cavity drain systems.
(c) Compartment and access arrangements.

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(d) Arrangement of tanks.
(e) Electrical generator location and numbering.
(f) Bulkheads (pressure).
(g) Hull and hull openings.
(h) Tanks and compartments (volumes).

3.5.6 System and component diagrams. Each installed system (piping, electrical, and electronic) shall be illustrated through the use of functional schematic diagrams (see figures 10-6 and 10-7). Where essential to the understanding of system operation, components of the system shall be separately illustrated and may be depicted using other types of drawings such as perspective, orthographic projection, cross-section, or isometric projection. Text shall be restricted to safety precautions and shall only be used to provide instructions or explanations which cannot be covered adequately by annotating the illustrations. Systems which shall be covered include, but are not limited to the following categories:

(a) High pressure air systems.
(b) Low pressure air systems.
(c) Ventilation systems.
(d) Gas (refrigerants, oxygen, compressed gasses, and so forth) distribution systems.
(e) Water (potable, sea water, air conditioning, chilled water for electronic system cooling, waste, drain, and so forth) systems.
(f) Hydraulic systems.
(g) Electric power [60 hertz (Hz) and 400 Hz, alternating current (a.c.), 250 volts direct current (Vdc), and battery] generation and distribution systems.
(h) Interior communication (IC) systems.
(i) Fire alarm, firefighting, and damage control systems.
(j) Steam systems.
(k) Feed and condensate systems.
(l) Distilling plant systems.
(m) Fuel and lube oil systems.
(n) Those systems presently covered by the engine room fluid systems composite.
(o) Salvage systems.
(p) Oxygen generator, CO_2 scrubber, CO-H_2 burner.
(q) Electronic equipment cooling flow.
(r) Escape systems.

3.5.6.1 Specific examples. Specific examples of system and component diagrams to be included are presented below. Note that the generic categories and titles under which the examples have been placed are neither exclusive nor all-encompassing.

VOLUME I - PIPING

Air Conditioning and Ventilation:

Air Conditioning Chilled Water System
Refrigerant 11 Air Conditioning Plant
Ventilation and Air Conditioning Control System

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Ventilation System
Ventilation Flow Diagram

Atmosphere Systems and Components:

Atmosphere Analyzing System
CO_2 Removal System
CO-H_2 Burner
Lithium Bromide Absorption System
Main Oxygen System
Oxygen Generator System

Compressed Air Systems and Components:

High Pressure Air System
HP Air System Air Dryer
HP Air Compressor Condensate Drain
700 lb/in^2 Service Air System
100 and 150 lb/in^2 Service Air Systems
20 lb/in^2 Service Air System
Diesel Engine Air Start

Emergency Systems:

Air Salvage System
Emergency Air Breathing System
Forward and After Escape Trunks

Fuel Oil Systems:

Diesel Generator Fuel System
Fuel and Compensating Water System
Fuel Filter Separator

Fresh Water Systems and Components:

Fresh Water Drain Collecting System
Plumbing System
Potable Water System
3000 Gallons per Day Distilling System
1800 Gallons per Day Distilling System
Auxiliary Fresh Water Cooling System
Diesel Generator Fresh Water Cooling System

Hydraulic Systems and Components:

Hydraulic Power Plant
Main and Vital Hydraulic System
External Hydraulic System
Snorkel Mast Hydraulic Operation
Periscope Hydraulic Operation
Antenna Masts Hydraulic Operation

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nance schematic diagrams in Chapter 5. Detailed

Hydraulic Accumulator Indicators
Steering Hydraulic System
Fairwater Planes Hydraulic System
Stern Planes Hydraulic System
Windlass and Capstan Hydraulic Operation
Steering and Diving Control Station

Lube Oil Systems and Components:

Diesel Generator Lub Oil System
Main Lub Oil System
Shaft Lube Oil System
SSTG Lube Oil System
Turbine Lube Oil Fill, Transfer, and Purification System

MBT Systems and Components:

MBT Flooding and Venting
MBT HP Blow System
MBT HP Blow Valves
MBT LP Blow System

Sea Water Systems and Components:

Main Sea Water Cooling System
Main and Auxiliary Sea Water Hull and Backup Valve Operation
Auxiliary Sea Water Cooling System
Air Conditioning Sea Water Cooling System
Diesel Generator Sea Water Cooling System
Drain System
Auxiliary Drain System
Gravity Drains
Surge Tank Arrangement

Snorkel System:

Snorkel System and Panels
Inboard and Outboard Induction and Ventilation Exhaust Valves
Induction Head Valve
Inboard and Outboard Diesel Exhaust Valves

Steam Systems and Components:

Auxiliary Steam System
Gland Seal and Exhaust Systems
High Pressure Steam Drain System
Steam Generator Cutout Valve Operation
Main Steam Root Valve and Crossover Valve Operation

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Trim System and Components:

Trim System Suction and Discharge Valves
Trim System - Forward and Aft
Hovering and Depth Control System

Miscellaneous Systems and Components:

Battery Electrolyte Agitation System
Clutch Control System
Depth Gage Piping
Floating Wire Antenna Operating Gear
Forward and After Signal Ejectors
Mode Selector Switch
Propulsion Turbine Throttle Valve and Governor Operation
Refrigeration System
SSTG Throttle Valve and Governor Operation
Torpedo Loading, Stowage and Handling
Torpedo Firing Sequence
Torpedo Tube Hydraulic System
Torpedo Tube Flood and Drain System
Trash Disposal Unit
Trash Disposal Compactor
Vibration Reducer Control System

VOLUME II - ELECTRIC - ELECTRONICS

Electric Propulsion:

Electric Propulsion Motor Control
Propulsion Clutch and Hydraulic Pump Control
Secondary Propulsion Motor Control
Storage Battery - Cell Arrangement and Metering

Electric Power Generation and Control:

Electric Power Sources and Major Circuit Breakers
Motor Generator Sets Operating Controls
Ship Service Turbine Generator Sets Operating Controls
Diesel Generator Control
Electric Power Generation Block Diagram
400 Hz Motor Generator Sets Control
Synchronizing Scheme Diagram
Electric Plant Circuit Breakers Control
Ground Detection

Electric Power Distribution and Lighting:

Ship Service 400 Hz Power Distribution
Outboard Lighting
Ship Service Lighting Power Distribution
Emergency Lighting Power Distribution

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Electrical Auxiliaries:

Lube Oil and Fuel Oil Equipment
Auxiliary Sea Water Cooling Pumps Equipment
Main Sea Water Cooling Pumps Equipment
Auxiliary Fresh Water Cooling Pumps Equipment
Main Feed Pumps Equipment
Main Condensate Pumps Equipment
Ventilation and Heating Equipment
Air Conditioning Equipment
High Pressure Air Compressor and Low Pressure Blower Equipment
Trim and Drain Pumps and Submersible Pumps Equipment
Hydraulic Power Plant Equipment
Distilling Equipment
Atmosphere Analyzer System
Hydrocarbon Analyzer System
Air Sampling and Analysis Equipment
Atmosphere Cleansing and Replenishment Equipment
Galley and Laundry Equipment
Refrigeration Equipment

Interior Communications:

Main Ballast Tank Vent Valves Indicators and Control System
Ballast Control Panel
Ballast Control Panel Common Alarm Circuits
Depth and Course Control System
Hovering and Depth Control System
Ship Control Panel
Pressure Indicating System
Snorkel System
Hull Opening Indicator and Control System
Trim and Drain Indicator and Control System
High Pressure Air Indicator and Control System
Electronic Mast Position and Hoist Control System
Remote Tank Level Indicator System
Trim and Drain Pump Remote Flow Indicator System
Refrigeration Temperature Indicator System
Hydraulic Accumulator Contents Indicator System
Rudder, Fairwater and Stern Planes Angle Indicator System
Depth Indicator System
Ship Safety Alarm System
Dead Reckoning System
Maneuvering Room Panel Common Alarm Circuits
Reactor Compartment Bilge Liquid Level Indicator and Alarm System
Propeller Shaft Revolution Indicator System
Engine Order and Wrong Direction Indicator System
Battery Electrolyte Agitation System
Battery Air Flow Indicator System
Hydrogen Detector System
Gyrocompass Repeater and Gyrocompass System
Underwater Log and Dummy Log System

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Ships Angle Indicator System
Trim Angle Indicator System
Main and Auxiliary Sea Water System Status Panel
Sea Water Valve Control and Position Indicator System
Salinity Indicator System
Temperature Monitoring System
Electronic Equipment Cooling Water Temperature and Flow Monitoring System
Propeller Shaft IC Circuit
Diesel Generator IC Circuit
Integrated Announcing System
Officer Call System
Dial Telephone System
Sound Powered Telephone System

Steam Plant:

Steam Plant Valve Position Indicator System
Steam Plant Alarms

Electronic Systems:

60 Hz Electronics Power Distribution
400 Hz Electronics Power Distribution
ECM Systems
Radar Systems:
- Surface Search
- Air Search
- Tracking
- IFF

Radio Systems:
- Communications
- Tactical
- LORAN
- SatNav
- RDF

Sonar Systems:
- Receiving
- Detecting Ranging
- Depth Sounding
- Distress Alert
- Underwater Telephone
- Tracking - Speed
- Hydrophone and Transducer Arrangement

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use

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his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality control. Unless otherwise specified (see 6.2), the quality conformance inspection, Government inspection, validation, verification, technical manual quality assurance (TMQA) program reviews, technical manual evaluation records and rejection requirements shall be in accordance with MIL-M-24784.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified (see 6.2), the packaging and preservation requirements for delivery of book plans, review draft copies, preliminary technical manuals, final reproducible copy, technical manuals, replenishment materials, changes and revisions shall be in accordance with MIL-M-24784.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. TABs prepared in accordance with this specification are primarily intended as a training aid for use by the ship crews in studying the installed systems of the ship. The TAB is also intended to be used as a reference document by engineering and technical personnel.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification (or any TMCR referencing this specification).
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).
(c) Applicable publication number, classification, and distribution statement (see 3.1).
(d) An outline or book plan manual must be specified whenever a technical manual is required for a new equipment or system (see 3.2a).

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(d) Test equipment setup and other illustrations necessary to s

(e) Double column text must normally be specified. When increased effectiveness of presentation results, a single column format is also acceptable.
(f) A review draft copy (RDC) must always be specified for every new technical manual (see 3.2b). An RDC also must be specified for manual change or revision procurements (see 3.2f and 3.2g), unless changes to the technical manual have been pre-approved (TMDER, ECP, and so forth) by the Government.
(g) Specify the number of weeks required for Government review of technical manual deliverables.
(h) Specify a preliminary technical manual when an extended period of use is required prior to Government verification (normally applies to complex equipment and systems only), see 3.2c.
(i) Final reproducible copy (see 3.2d and 3.2.1).
(j) Supplements to manuals (see 3.2e) must be specified when:

(1) Classified information can be confined to the supplement manual such that its basic technical manual will be routed as "unclassified".
(2) It augments a technical manual to provide for a different model and can be justified as a cost effective method of promulgation.
(3) It can be justified as a practical method of promulgation.

(k) Changes must be specified only when they pertain to technical manual corrections resulting from a hardware change or modification. They may also be specified if it has been determined that significant omissions of technical data or information has been identified. Do not specify a change to correct minor or editorial corrections. When changes are justified on the above basis, the change package must include the correction of all outstanding temporary or interim changes and may include minor changes (see 3.2f). The change package must be developed in accordance with MIL-M-24784/1.
(l) Whenever the number of change pages is estimated to exceed approximately 25 percent of the total number of pages in the manual, an update revision must be specified (see 3.2g) and developed in accordance with MIL-M-24784/2.
(m) When the total number of pages requiring change exceed approximately 50 percent of the total number of pages in the manual, a complete revision must be specified (see 3.2g) and developed in accordance with MIL-M-24784/2.
(n) What quality assurance and management data items are to be delivered (see 3.2h, 3.2.4, 4.2, 6.3 and MIL-M-24784).

(1) TMQA program plan.
(2) Validation plan.
(3) Validation certification.
(4) Evaluation records.
(5) Verification plan.
(6) Verification incorporation certification.

(o) Typography, if other than specified (see 3.2.1.2).

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Long term requirements shall be requested at least 3 months prior to submittal of

(p) Color in illustrations, if other than specified (see 3.4.4.7 and MIL-M-24784).
(q) When preservation, packing, and marking are other than as specified (see Section 5. PACKAGING).
(r) For classified equipment, attach a DD Form 254 specifying the following:

(1) Security classification.
(2) Downgrading and declassification notification.
(3) Areas requiring security protection.

6.2.1 Distribution statement. A statement will be provided by the Government defining the conditions of availability for distribution, release, or disclosure of the data contained in the TAB. This statement must be included on the cover and title page in the appropriate location.

6.2.2 Page size. Only when the increased cost of production can be justified by the effectiveness of the page layout will the alternate page size of 4-1/4 by 11 inches be specified. When this alternate is specified, the acquisition documents must prohibit the use of foldouts and foldups (see 3.2.1.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
3.2.4	DI-M-2194	Manual, Technical: Quality Assurance Program Plan
3.2.4	DI-M-2195	Manual, Technical: Validation Plan
3.2.4	DI-M-2196	Manual, Technical: Validation Certification
3.2.4	DI-M-2197	Manual, Technical: Evaluation Record
3.2.4	DI-M-2198	Manual, Technical: Verification Plan
3.2.4	DI-M-2201	Technical Manual Verification Incorporation Certification

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Definitions. The words or phrases used throughout this specification are defined in MIL-M-24784.

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6.5 Subject term (key word) listing.

Equipment
Pictorial
Schematic Systems

Review activity:
Navy - EC

Preparing activity:
Navy - SH
(Project TMSS-N242-10)

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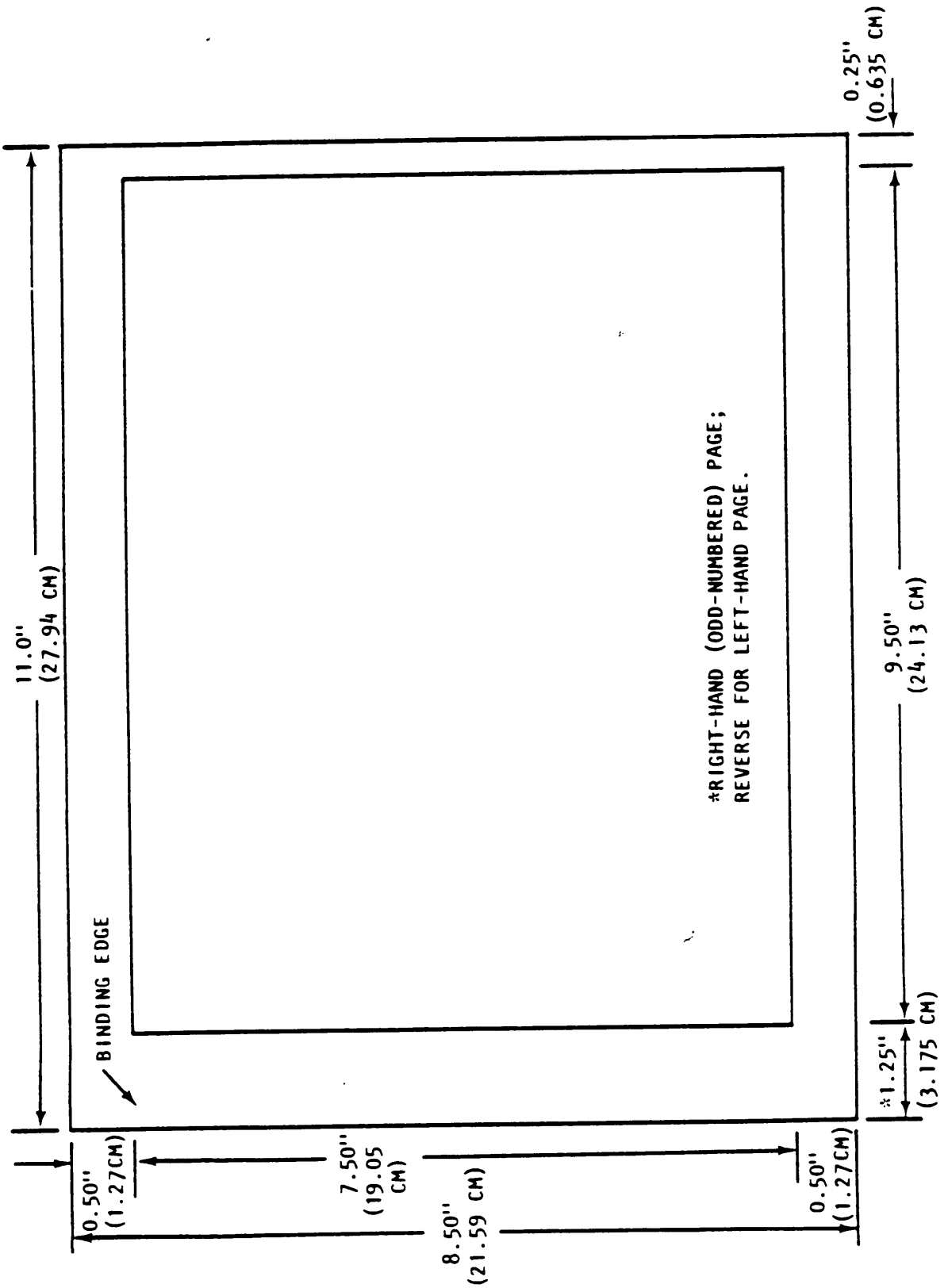


FIGURE 10-1. Page size and format.

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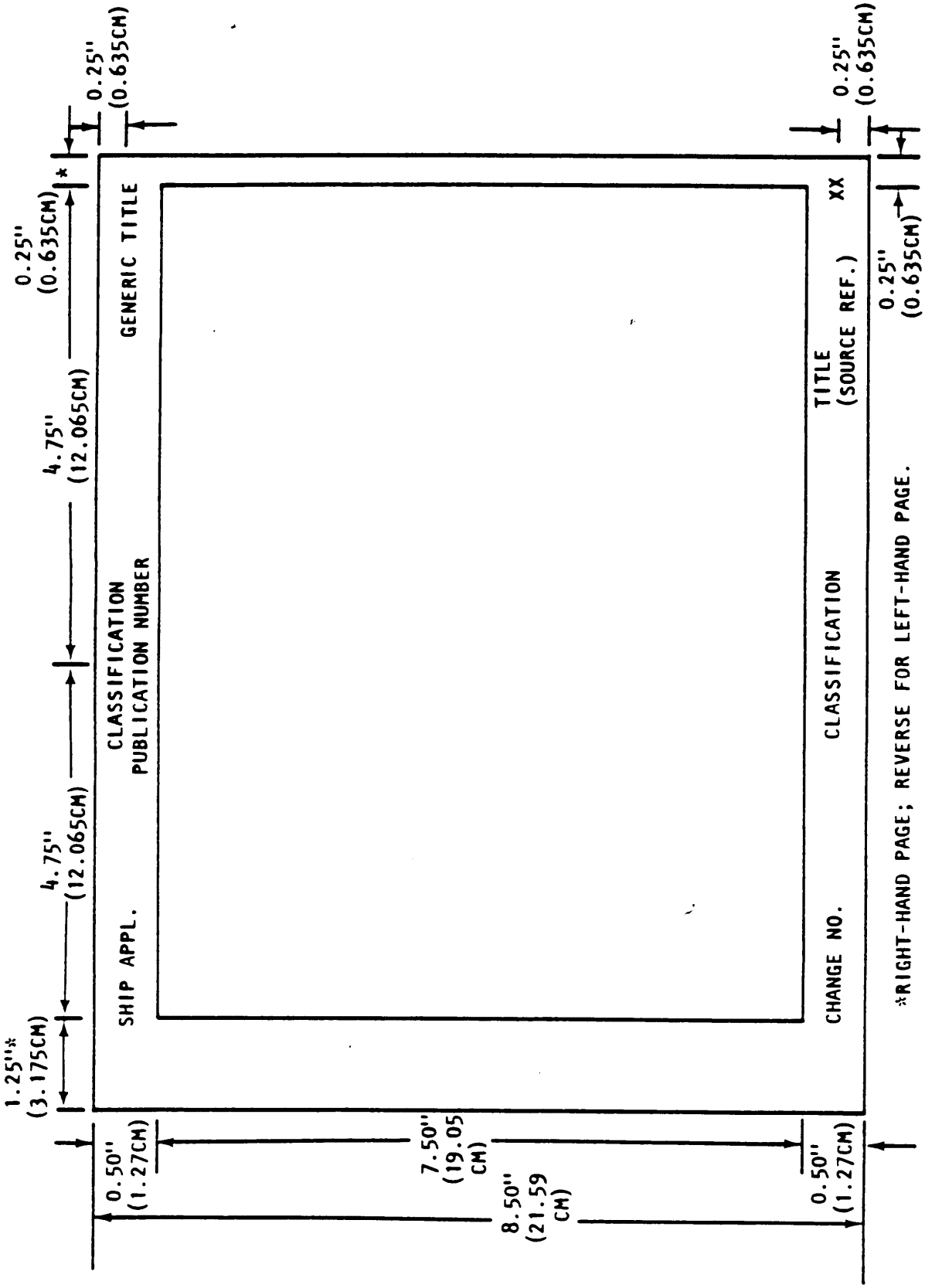


FIGURE 10-2. Single illustration page layout and dimensions.

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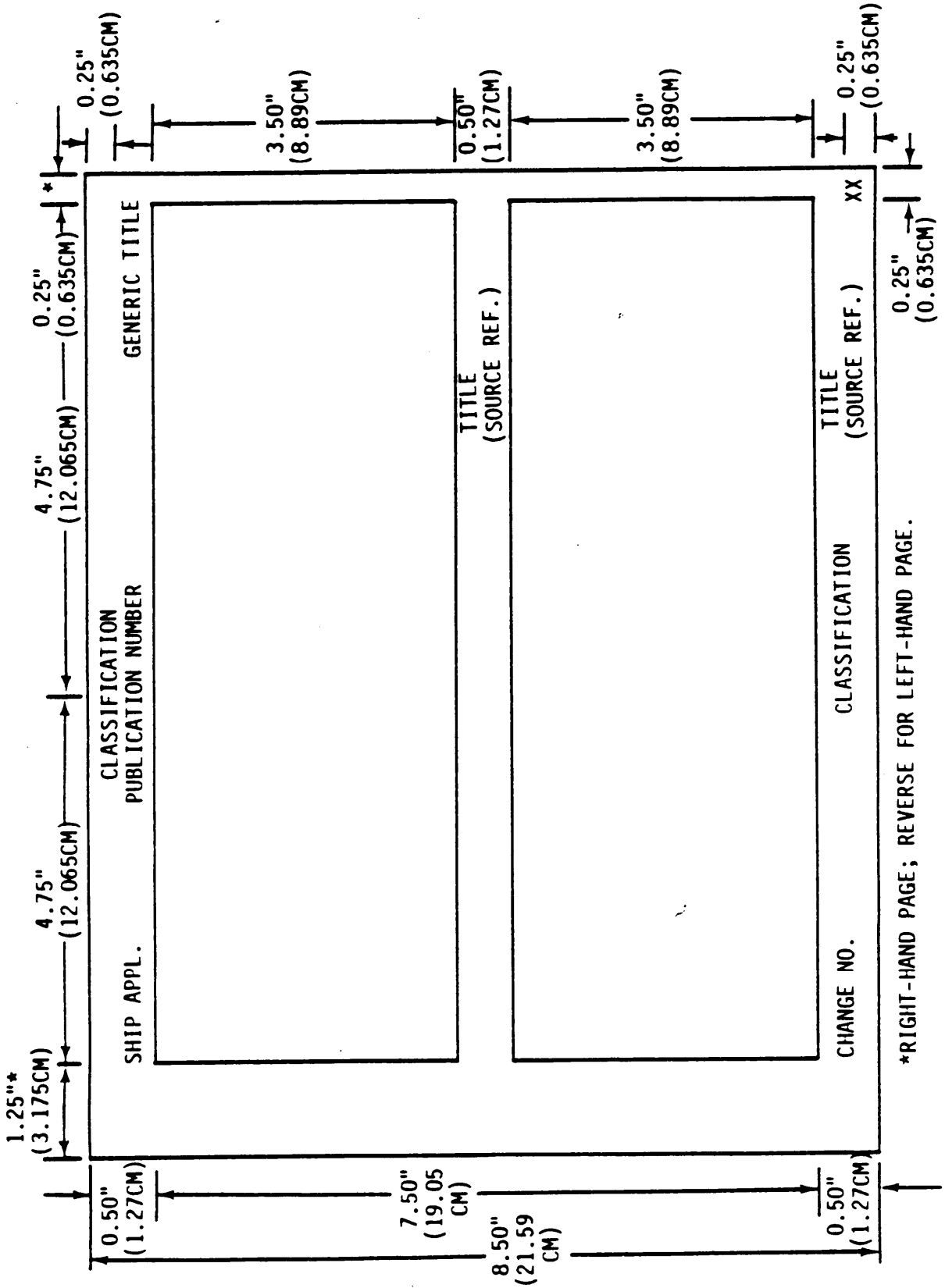
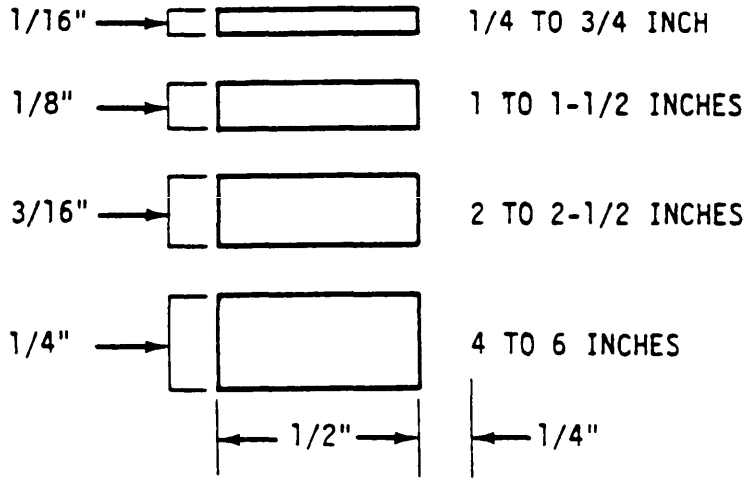


FIGURE 10-3. Double illustration page layout and dimensions.

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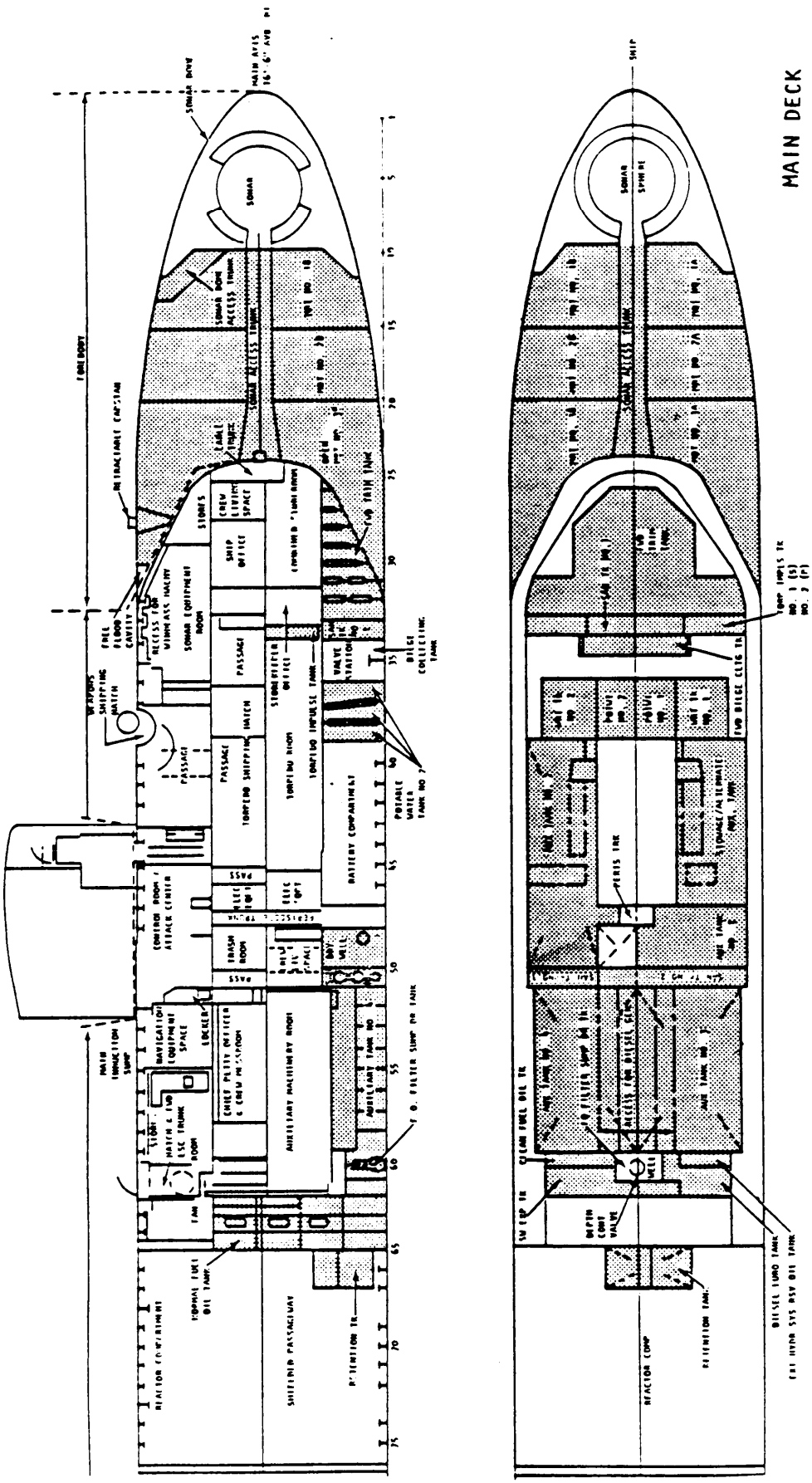


NOTE

PIPE SIZING IS SUBJECT TO VARY, AS IT IS DEVELOPED FROM THE PIPE SIZES THAT ARE INDICATED ON THE SYSTEM DIAGRAM BEING USED.

FIGURE 10-4. Recommended pipe size scale.

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NOTE: Sample shows arrangement only. Does not meet minimum legibility requirements.

FIGURE 10-5. Plan view. (Sample)

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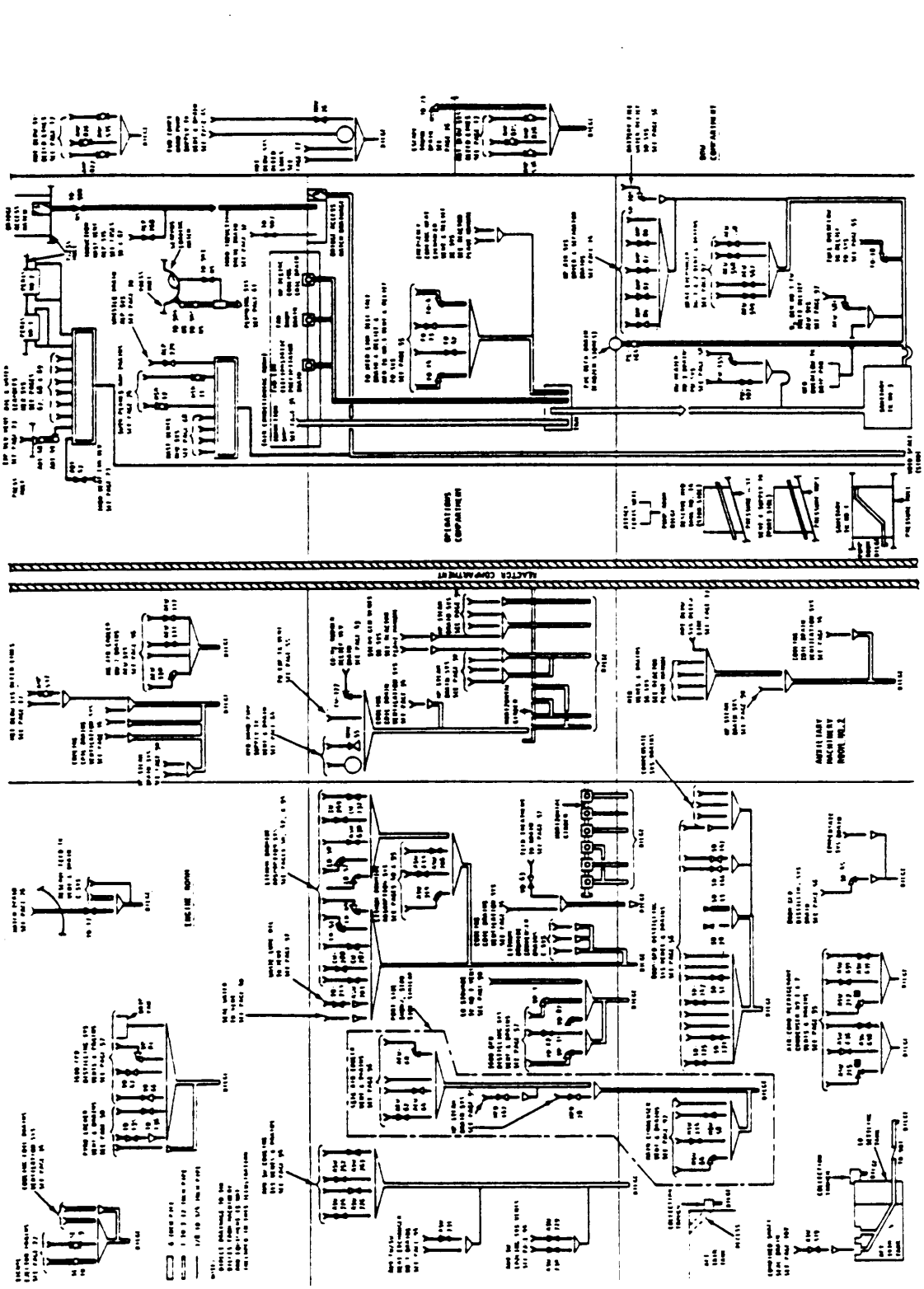
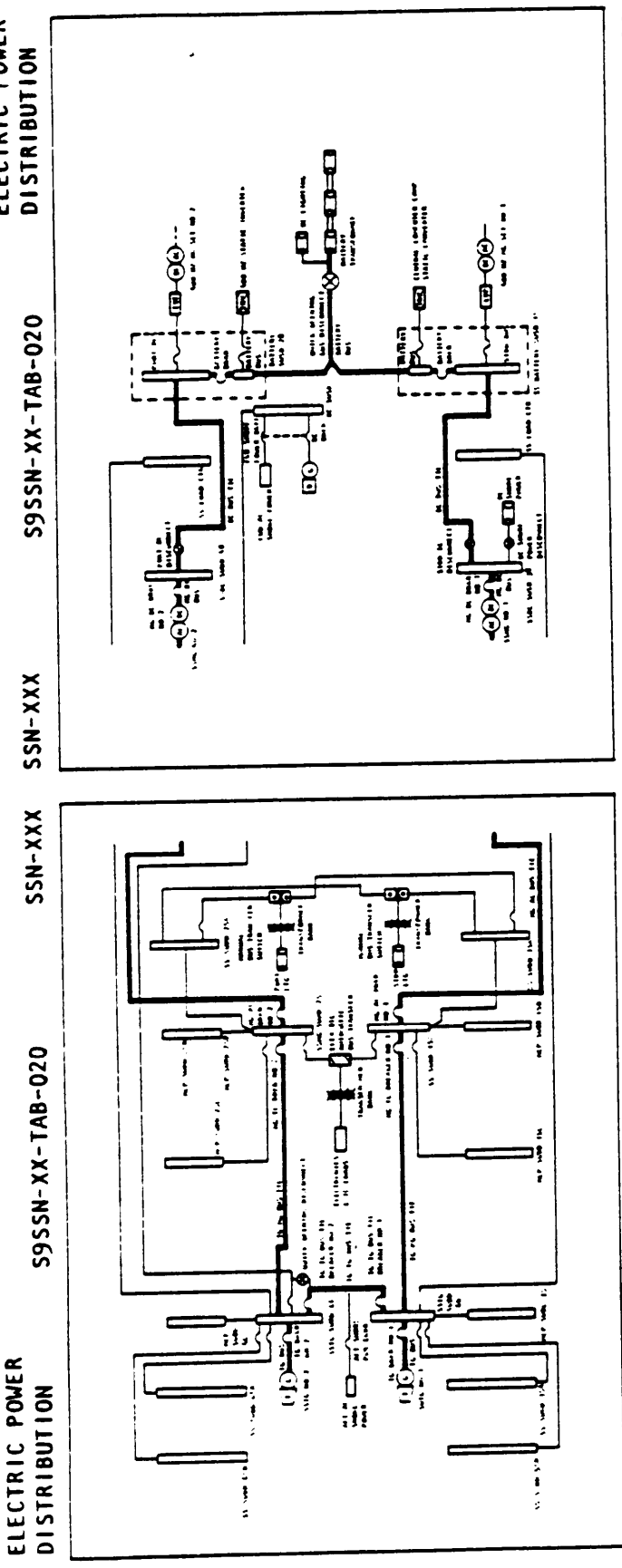


FIGURE 10-6. Functional schematic diagram. (Sample)

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NOTE: Sample shows arrangement only. Does not meet minimum legibility requirements.

FIGURE 10-7. Functional schematic diagram, 2-page spread. (Sample)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-24784/10(SH)	2. DOCUMENT DATE (YYMMDD) 940906

3. DOCUMENT TITLE
MANUAL, TRAINING AID BOOKLET (TAB) REQUIREMENTS FOR NAVAL SHIPS SYSTEMS

4. NATURE OF CHANGE (identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

A. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

A. NAME Technical Point of Contact (TPOC) MR. MICKEY ANDER, SEA 04TD3 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: TPOC: 703-602-8701 DSN: 8-332-8701
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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